SurfPrior-GC: a reproducible framework for uncertainty-aware gastric cancer surface-target prioritization with GPI evidence-routing audit

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Abstract

We introduce SurfPrior-GC, a reproducible framework for gastric cancer surface-target prioritization that integrates heterogeneous public evidence while keeping uncertainty explicit rather than collapsing it into a single ranked list. The central methodological finding is a GPI-anchor evidence-routing audit: topology-first routing had excluded 54 confirmed glycosylphosphatidylinositol (GPI)-anchor genes from the ranked universe. Crediting confirmed lipidation evidence at universe construction increased the ranked Core+Probable surfaceome from 2,650 to 2,704 genes, raised confirmed GPI-anchor representation from 65 to 119 genes, moved *CEACAM5* from rank 120 to 12 and *MSLN* from rank 453 to 158, and produced a median improvement of 453 positions among common confirmed GPI-anchor genes. SurfPrior-GC then integrates tumor/normal RNA sequencing, Human Protein Atlas (HPA) protein and localization evidence, UniProt topology and lipidation annotations, curated surfaceome resources, benchmark controls, and stability analysis. The final output was a coarse, unordered set of experimentally testable Tier 1/2 hypotheses, including six Tier 1 candidates, with explicit caveats for bulk RNA, compartment attribution, isoform resolution, HPA immunohistochemistry granularity, and absence of wet-lab validation. Tier 1/2 nominations were consistent with contemporary surface-target resources, but their overlap was not enriched beyond a matched null controlling for surfaceome confidence, expression, protein evidence, topology, component completeness, and accessibility. SurfPrior-GC supports reproducible,

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transparent hypothesis generation with explicit uncertainty boundaries while making public bulk-data limits part of the result.

Keywords: gastric cancer, surfaceome, target prioritization, reproducibility, GPI-anchored proteins, proteogenomics

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1. Introduction

1.1. Why surface-target prioritization is hard

Surface-target prioritization from public data is not equivalent to ranking genes by tumor RNA abundance. RNA expression does not establish protein presence, membrane localization, extracellular accessibility, tumor-cell attribution, isoform specificity, or an acceptable normal-tissue profile (Uhlen et al., 2015, 2017; Thul et al., 2017; Moek et al., 2017; Li et al., 2024). Bulk tumor RNA can reflect tumor microenvironment (TME) compartments, including endothelial, immune, and stromal cells, rather than malignant epithelial cells (Yoshihara et al., 2013; Han et al., 2023; Zhang et al., 2019; Jeong et al., 2021). Surfaceome resources also differ in how they route non-transmembrane evidence, including glycosylphosphatidylinositol (GPI)-anchored proteins; prior in silico surfaceome work explicitly accounted for annotated GPI-linked proteins, so the open problem here is not whether GPI anchors can be surface evidence but how incomplete evidence routing propagates into an auditable candidate universe and ranking (Bausch-Fluck et al., 2018).

1.2. Clinical context and contribution

In gastric adenocarcinoma, antibody-accessible surface proteins remain a tractable intervention space because they can support antibody-based therapeutics, antibody-drug conjugates (ADCs), cell therapies, receptor blockade, or imaging strategies (Bray et al., 2024; Cancer Genome Atlas Research Network, 2014; Hu et al., 2021; Bausch-Fluck et al., 2015, 2018; Li et al., 2022; Razzaghdoust et al., 2021; Kathad et al., 2024; Moek et al., 2017; Li et al., 2026; Katoh et al., 2025). Established and emerging gastric cancer targets, including ERBB2, CLDN18.2, FGFR2b, TACSTD2/TROP2, EPCAM, CEACAM5, MET, MSLN, CDH17, CD66c, and PVR/CD155, provide useful benchmarks but do not remove the need for systematic prioritization (Bang et al., 2010; Shitara et al., 2023; Shah et al., 2023; Wainberg et al., 2022;

Xu et al., 2025; Chang et al., 2026; Zhang et al., 2025; Zhao et al., 2026; Li et al., 2026; Katoh et al., 2025; Wang et al., 2025).

Recent literature already includes pan-cancer surfaceome resources, pan-cancer ADC target discovery, tumor-specificity modeling, gastric clinical-trial target prioritization, gastric proteogenomic vulnerability atlases, and single-target or subtype-focused gastric/GEJ/GSRCC ADC discovery studies (Hu et al., 2021; Li et al., 2022; Razzaghdoust et al., 2021; Kathad et al., 2024; Wang et al., 2026; Deng et al., 2024; Moek et al., 2017; Li et al., 2024; Xu et al., 2025; Chang et al., 2026; Zhang et al., 2025; Zhao et al., 2026; Li et al., 2026; Katoh et al., 2025; Wang et al., 2025). We therefore introduce SurfPrior-GC, a release-grade, uncertainty-aware framework for auditing how heterogeneous public surfaceome evidence is combined, bounded, and converted into experimentally testable hypotheses in gastric cancer. Its main transferable contribution is a GPI-anchor evidence-routing audit: SurfPrior-GC quantifies how topology-first routing excluded 54 confirmed GPI-anchor genes, corrects this at universe construction before scoring, and preserves the downstream effect on ranking and benchmark behavior. The framework then integrates RNA, protein/localization, topology, normal-risk, TME, benchmark-control, and stability evidence into a coarse experimental queue. External TCSA and Wang comparisons are used as consistency and scope checks, with matched-null behavior reported as an evidential boundary rather than candidate-level experimental evidence.

2. Material and methods

2.1. Reproducibility design

The analysis was implemented as a staged reproducible workflow rather than as an exploratory notebook. Controls, exclusion criteria, dataset targets, scoring scenarios, random seeds, tissue mappings, and coarse tiering rules were pre-specified in repository configuration and decision files before candidate interpretation; this is an operational repository freeze, not a formal external registry entry. Raw or downloaded inputs were checksum-registered where stored locally, and each major analysis stage produced both machine-readable tables and a short report. The repository preserves design decisions, analytical decisions, provenance records, a release manifest, and stage-specific validation checks, following reproducible-computational-research and findable, accessible, interoperable, reusable data principles where feasible (Sandve et al., 2013; Wilkinson et al., 2016).

Workflow integrity was assessed with three levels of checks. First, unit tests cover core scripts and stage-specific invariants. Second, a smoke test verifies expected outputs from data inventory through manuscript figure/table packaging, including manual curation artifacts. Third, Snakemake declares the workflow outputs and reports whether any declared artifact is missing or out of date. Some early bootstrap and data-inventory outputs were generated before all rules were formalized and therefore retain missing Snakemake provenance metadata; these steps write source inventories and checksum context rather than scoring outputs, and all later analysis, ranking, and manuscript-package outputs are tracked.

Intermediate ranking snapshots were retained rather than overwritten. The pre-fix snapshot preserves the ordinal surfaceome-confidence transform, the pre-GPI snapshot preserves the state before the GPI-anchor correction, and the final frozen ranking snapshot preserves the active ranking after the universe-wide GPI evidence correction and downstream rerun. This version history was kept to document bug detection and correction, not to select a favorable ranking retrospectively.

2.2. Datasets and identifier normalization

The main input sources are summarized in Table 1. TCGA-STAD and GTEx expression data were obtained through the UCSC Xena/Toil RNA-seq recompute and used as the primary tumor-normal RNA matrix (Goldman et al., 2020; Vivian et al., 2017; GTEx Consortium, 2020). GDC TCGA-STAD metadata and cBioPortal TCGA-STAD clinical and copy-number fields were used for clinical covariate availability, selected amplification context, and secondary sensitivity planning (Grossman et al., 2016; Cerami et al., 2012). Human Protein Atlas (HPA) downloadable files supplied normal IHC, cancer IHC, subcellular localization, and tissue RNA context (Uhlen et al., 2015, 2017; Thul et al., 2017). UniProtKB reviewed human records supplied protein identifiers, topology, transmembrane annotations, extracellular features, signal peptides, lipidation/GPI-anchor evidence, protein length, and isoform mappings (UniProt Consortium, 2025). TCSA, CSPA, SURFY, UniProt GO, UniProt topology/GPI fields, and HPA subcellular localization were integrated to construct the surfaceome universe (Hu et al., 2021; Bausch-Fluck et al., 2015, 2018).

Identifier normalization used HGNC-approved protein-coding genes as the primary denominator before surfaceome filtering (Tweedie et al., 2021).

This choice prevents the candidate denominator from being inflated by non-coding or deprecated matrix rows while preserving alias, Ensembl, UniProt, and HPA mappings for evidence integration. Mapping failures and source coverage are retained as audit outputs rather than silently dropped.

2.3. Surfaceome universe construction

The surfaceome universe was defined at the HGNC gene level. Core/Probable membership required independent surface support plus an anchor, topology, or localization signal. Qualifying anchor evidence included UniProt extracellular topology, UniProt transmembrane annotation, confirmed UniProt lipidation GPI-anchor, HPA plasma membrane localization, or SURFY surfaceome support. Curated-list or GO-only genes lacking such support were retained as ambiguous surface-context genes rather than treated as high-confidence surface targets.

The final Core+Probable universe contains 2,704 genes: 2,646 Core and 58 Probable. Published-list overlap was used as a sanity check, with Jaccard overlap of 0.7749 for TCSA, 0.2889 for CSPA, and 0.8017 for SURFY. Negative controls did not enter Core/Probable, and positive/benchmark controls were present or documented.

Confirmed UniProt lipidation GPI-anchor evidence was credited as a non-experimental strong anchor signal at the universe construction stage. Specifically, confirmed GPI evidence contributed score +2, one support source, `has_anchor=true`, and `has_strong=true`. Subcellular-location-only GPI annotations were flagged but not credited as confirmed direct lipidation evidence. This universe-wide correction was applied before final ranking rather than as a target-specific patch.

2.4. Tumor expression and heterogeneity

Tumor expression was computed for the Core+Probable universe using TCGA-STAD primary tumor samples from the Xena/Toil matrix. Xena/Toil values were treated as $\log_2(\mathrm{TPM} + 0.001)$, transformed back to TPM as $2^x - 0.001$, and clipped at zero. Of 2,704 Core+Probable genes, 2,696 had measured Xena expression and 8 were missing.

The tumor expression component E combines abundance and prevalence. For each measured gene, percentile ranks were computed for median tumor TPM, the fraction of tumor samples with TPM > 1, P75 tumor TPM, and P90 tumor TPM. The raw expression score was:

$$
\begin{aligned}
E_{\text{raw}} = 0.40\, r_{\text{median}} + 0.30\, r_{\text{prev}>1} \\
+ 0.20\, r_{\text{P75}} + 0.10\, r_{\text{P90}}.
\end{aligned} \tag{1}
$$

The E score is a component score only. It is not by itself a final target ranking.

Molecular subtype, stage, tissue-origin, and selected amplification summaries were computed when fields were available. TCGA molecular subtypes were used for expression and power summaries. Exact Lauren subtype was not available in the current raw sources; histology proxy fields were retained only as audit information and were not treated as quantitative Lauren labels. cBioPortal GISTIC calls were queried for ERBB2, FGFR2, and MET to provide amplification context.

2.5. Normal selectivity and off-tumor risk

Normal selectivity and on-target/off-tumor risk were estimated with the same TPM transformation as tumor expression. TCGA-STAD primary tumor, GTEx stomach normal, and TCGA-STAD adjacent normal samples were summarized separately. The primary stomach normal comparator used GTEx stomach because the adjacent-normal set was source-matched but smaller.

The selectivity component N combines two tumor-normal contrasts and a statistical rule. N_{stomach} compares tumor median TPM with GTEx stomach median TPM. N_{critical} compares tumor median TPM with the maximum mapped critical normal expression across Xena/HPA-derived normal tissue contexts. The component score was:

$$
\begin{aligned}
N_{\text{score}} = 0.50\, r_{\text{critical}} + 0.30\, r_{\text{stomach}} \\
+ 0.20\, N_{\text{stat,GTEx}}.
\end{aligned} \tag{2}
$$

The pre-specified positive statistical rule required tumor versus GTEx stomach log2 fold-change >= 1 and BH-FDR < 0.05. The TCGA adjacent-normal comparison was reported as a sensitivity context rather than as a full replacement for GTEx stomach.

The off-tumor risk component R used a pre-specified maximum weighted organ penalty. Critical organs and tissue classes were assigned manually chosen expert-prior weights, including heart and brain at 1.00, liver and kidney at 0.90, lung and endothelial contexts at 0.85, hematopoietic at 0.80, immune at 0.75, gastrointestinal epithelial at 0.70, and reproductive/other at 0.60. These weights are not learned, optimized, or inferred from outcome data. R is high-worse: higher values indicate greater on-target/off-tumor concern and are subtracted in the integrated score. Alternative $R_{\text{max+breadth}}$ and $R_{\text{sum,capped}}$ forms were retained for sensitivity analysis.

2.6. Protein and localization evidence

Protein/localization evidence was computed from HPA stomach cancer IHC, HPA normal IHC, and HPA subcellular localization. In the 2,704-gene Core+Probable universe, HPA stomach cancer IHC covered 1,790 genes, HPA normal stomach and critical normal IHC covered 1,535 genes, and HPA subcellular localization covered 1,302 genes. Membrane or cell-junction localization support was present for 734 genes. CPTAC/PDC proteomics was not assessed in the primary analysis and was not imputed.

The P component combines tumor protein presence, membrane or cell-junction localization support, mapped normal protein safety support, HPA reliability support, and discordance penalties. The HPA bulk cancer file does not expose antibody IDs, staining intensity/quantity fields, patient-level membrane pattern, or multi-antibody concordance. These unavailable fields were not inferred and are marked for candidate-card review where relevant.

Discordance flags include HPA-missing stomach cancer IHC, high RNA but absent tumor protein, protein present without membrane/cell-junction support, and low-confidence HPA evidence. HPA absence does not automatically eliminate a candidate because the protein layer has incomplete coverage and antibody-level detail is unavailable in the bulk files.

2.7. Tumor microenvironment specificity

No processed gastric single-cell RNA-seq dataset was admitted into the numeric primary score. The optional SC component therefore remains `not_available` and has zero weight in the primary ranking.

As a pre-specified fallback, bulk TCGA-STAD expression was used to flag possible TME-derived signal. Marker modules were computed for CAF/fibroblast, endothelial, myeloid/macrophage, T-cell, and B/plasma compartments. For each candidate gene, Spearman correlations were computed between candidate expression and each module score across 414 TCGA-STAD tumors. ESTIMATE stromal and immune signatures from `tidyestimate` were also computed from the same Xena/Toil matrix and used as relative purity/admixture covariates for partial Spearman correlations (Yoshihara et al., 2013).

After the ranking and tiers were frozen, processed TISCH2 gastric cancer summaries were used only as candidate-level compartment annotations for the 18 Tier 1/2 genes (Han et al., 2023; Zhang et al., 2019; Jeong et al., 2021). `STAD_GSE134520` included a malignant-cell class, whereas `STAD_GSE167297` provided epithelial/TME context only. These annotations did not alter SC, the integrated score, or tier assignments.

These outputs are flags only. They are not hard ranking filters and are not interpreted as absolute pathology purity estimates. ESTIMATE and the TME marker modules both represent stromal/immune biology, so their partial correlations are biologically collinear rather than clean causal decomposition. A `high_purity_adjusted_tme_correlation` flag therefore triggers conservative review with protein/localization and cell-resolved evidence where available.

2.8. Topology, extracellular accessibility, and isoforms

Topology and extracellular accessibility were computed from reviewed UniProt human feature fields for all 2,704 Core+Probable genes. Parsed fields included protein length, topological domains, transmembrane segments, signal peptides, lipid/GPI anchor features, glycosylation, disulfide bonds, chains/domains, subcellular location, PTM comments, and Ensembl isoform mappings.

Each protein was assigned an accessibility class from A to E. Classes A/B indicate clear or inferred extracellular regions suitable for antibody-accessibility review. Class C indicates possible but less straightforward access, often for multipass proteins or shorter extracellular loops. Classes D/E are not automatic exclusions from the universe but block Tier 1 antibody-targeting claims unless later structure, literature, or candidate-card evidence supports an exception.

The topology component T combines accessibility class, extracellular length, topology confidence, and isoform confidence, then subtracts conservative cleavage/shedding and soluble-decoy penalties. These penalties are candidate-card flags, not hard exclusions. GPI-anchor features are explicitly captured in this layer; MSLN is the benchmark example where GPI anchoring supports membrane accessibility despite absence of a classical transmembrane segment.

Isoform-specific benchmarks were handled conservatively. *CLDN18* and *FGFR2* remain gene-level in the current expression layers, so CLDN18.2 and FGFR2b/IIIb claims are marked `isoform_unresolved` and are not used as evidence that the pipeline resolves isoform-specific expression.

2.9. Integrated scoring and sensitivity

The integrated score used the pre-specified components listed in Table 2. The primary scenario used manually chosen expert-prior weights $Surf = 0.15$, $E = 0.20$, $N = 0.20$, $R = 0.20$, $P = 0.15$, $SC = 0.00$, and $T = 0.10$. These weights are transparent design choices, not learned coefficients or optimal estimates. SC remained unavailable and was not imputed. Sensitivity scenarios included safety-prioritized, ADC-focused, novelty-focused, and protein-evidence-focused weighting; these were reported as sensitivity contexts, not alternative primary rankings.

$Surf$ is the relative surfaceome confidence within the admitted Core+Probable universe. After the GPI correction, it is scaled as:

$$\mathrm{Surf}_{\mathrm{relative}} = \frac{S_{\mathrm{conf}} - 5}{5}. \tag{3}$$

over the theoretical admitted confidence range 5 to 10, not over the observed min/max of the current dataset. This means that zero indicates the lowest admitted confidence level, not absence from the surfaceome universe.

Let x_{gj} denote the scaled component value for gene `g` and component `j`, w_j the primary weight, and A_g the available nonzero-weight components for that gene. The integrated score was:

$$S_g = \frac{\sum_{j \in A_g,\ j \neq R} w_j x_{gj} - w_R x_{gR}}{\sum_{j \in A_g} |w_j|}. \tag{4}$$

For score computation, missing principal components were handled with exclude-and-renormalize: the weighted score was computed over available nonzero-weight components and divided by the available absolute weight sum. Missingness was then carried into tiering restrictions and sensitivity analyses rather than mean-imputed. R is the only high-worse primary component and is subtracted from the total score. Rank percentiles used average-rank tie handling so tied raw component values received identical percentile values rather than HGNC-order-dependent ranks.

The stability analysis evaluated weight perturbation, leave-one-layer-out sensitivity, missing-data sensitivity, risk-threshold sensitivity, organ-weight perturbation, scenario stability, and post-scoring rank resolution. These analyses supported coarse tier language but not fine intra-tier ordering.

2.10. Tiering and manual curation

Tiering was performed only after stability analysis. The original fine tiering plan was collapsed to unordered coarse categories because post-scoring rank resolution supported coarse interpretation but not fine rank distinctions. The tiering rule file was frozen before candidate assignment. Tier 1 required stability, no unexplained critical normal risk, no unsupported TME-marker status, acceptable accessibility class, protein evidence or documented absence, no more than two missing primary components, and absence of single-layer dependence. Tier 2 captured candidates failing one Tier 1 criterion or showing scenario/subtype dependence. Watchlist captured candidates with major missingness, TME-marker behavior, unresolved benchmark isoforms, or other non-nominable but biologically relevant signals.

Manual curation informed tiers and caveats but did not change scores. The top 30 candidates were reviewed against UniProt, HPA, literature, clinical/druggability context, Wang 2026 concordance, and analysis-stage flags. Candidate notes explicitly record whether curation changed the score; all current entries preserve `changes_score=no`.

External baseline behavior was assessed in two ways. First, final scores were compared with the Cancer Surfaceome Atlas final and core GESP scores (Hu et al., 2021) over overlapping genes by Spearman correlation and top-k overlap. This is a pan-cancer surfaceome-prioritization baseline, not a gastric-specific validation label. Second, external consistency was assessed against Wang 2026 gastric proteogenomic surface-target resources (Wang et al., 2026). Two null models were retained for the Wang comparison. The first was a one-sided hypergeometric test against random draws from the frozen Core+Probable surfaceome universe. The second was a deterministic matched-null sensitivity analysis with 20,000 permutations and seed 20260531, sampling nearest-neighbor Core+Probable controls matched on surfaceome confidence, tumor expression percentile, protein-evidence percentile and missingness, topology percentile, available component count, and accessibility class. This cross-check is used as benchmark concordance and compartment-framework support, not as clinical-efficacy evidence or as per-gene single-cell support for genes absent from the Wang Figure 7H panel.

3. Results

3.1. SurfPrior-GC workflow and GPI-routing audit

SurfPrior-GC integrates public transcriptomic, protein, localization, topology, and curated surfaceome resources into a staged candidate-prioritization pipeline (Fig. 1; Table 1). The current execution covers dataset inventory, raw/source checksum capture, batch diagnostics, identifier normalization, surfaceome universe construction, tumor expression, normal selectivity and off-tumor risk, HPA protein/localization evidence, TME specificity flags, topology/isoform annotation, integrated scoring, stability analysis, coarse tiering, external baseline comparison, candidate-level scRNA compartment annotation, and manuscript figure/table packaging. Unit tests, smoke tests through the figure/table package, the manuscript brief check, and Snakemake dry-run checks pass for declared current outputs.

The final surfaceome universe contains 2,704 Core+Probable genes, comprising 2,646 Core and 58 Probable candidates (Fig. 2). The construction rule balanced published surfaceome support with direct anchor, topology, or localization evidence. This conservative rule excluded all intracellular/secreted negative controls from Core+Probable while retaining benchmark surface-target genes for downstream scoring.

During scoring diagnostics, confirmed GPI-anchored candidates exposed a source-integration problem. Treating GPI evidence only as a downstream topology note under-credited confirmed GPI-anchored surface proteins at the surfaceome universe layer. Prior surfaceome work already recognized annotated GPI-linked proteins (Bausch-Fluck et al., 2018); the issue quantified here is local evidence routing in a gastric cancer prioritization workflow. Confirmed lipidation evidence was therefore credited universe-wide as non-experimental strong anchor evidence before rerunning downstream analyses, while subcellular-location-only GPI annotations remained flagged but uncredited as direct lipidation evidence.

After correction, the active final ranking uses the fixed theoretical $Surf_{\text{relative confidence}}$ scale over the admitted confidence range 5 to 10. This means that low $Surf$ values represent lower relative confidence within an admitted surfaceome universe, not absence from the surfaceome. The GPI finding is therefore interpreted as a transferable under-crediting issue in resource integration, not as evidence that all GPI-anchored proteins are strong therapeutic candidates.

The impact of the correction was quantified against the preserved pre-GPI snapshot. The ranked universe increased from 2,650 to 2,704 genes; all 54 newly ranked genes were confirmed UniProt GPI-anchor genes. Confirmed GPI anchors increased from 65 to 119 ranked genes. None were in the pre-GPI top 50, whereas five confirmed GPI anchors entered the final top 50 (*BST2*, *CEACAM5*, *ALPG*, *ULBP2*, and *NT5E*), with zero classified as suspicious Surf-dominant entrants by the pre-stability audit. Among common GPI genes, the median rank improvement was 453 positions. The correction also changed benchmark behavior: *CEACAM5* moved from rank 120 to 12, and *MSLN* moved from rank 453 to 158. Because tiering was performed after the correction, no pre/post tier-change claim is made; instead, the current Tier 1/2 set contains three confirmed GPI candidates (*CEACAM5*, *BST2*, and *ALPG*).

3.2. External consistency and scope boundary

The primary behavior of SurfPrior-GC is transparent hypothesis generation with explicit uncertainty. Against the TCSA final GESP baseline, final scores showed moderate concordance across 2,685 overlapping genes (Spearman rho=0.389861, p=3.5e-98), with low top20 identity (1/20: *NECTIN2*) and 6 of 18 Tier 1/2 genes in the TCSA final-score top100. The TCSA core GESP baseline gave similar moderate concordance (rho=0.329466, p=5.3e-69), 2/20 top20 overlap (*ERBB2* and *ITGB4*), and 8 of 18 Tier 1/2 genes in the baseline top100. This supports external surfaceome consistency while showing that the final ranking is not a clone of a pan-cancer surfaceome score.

External consistency was also assessed against Wang 2026. Sixteen of eighteen Tier 1/2 nominations were present in Wang's drug-target table, including all six Tier 1 genes. Against the 2,704-gene Core+Probable universe, this exceeded a simple random draw (one-sided hypergeometric p=0.0013; odds ratio 7.27). However, a matched-null sensitivity analysis showed that the observed Wang overlap was not above matched expectation after controlling for surfaceome confidence, expression, protein-evidence availability, topology, component completeness, and accessibility class. For the all-table background, the matched-null mean overlap was 15.18 of 18 (upper-tail p=0.436); for the Wang membrane-protein flag, the matched-null mean was 13.99 of 18 (p=0.817). The external checks therefore support plausibility and benchmark consistency while leaving candidate-level experimental evidence to follow-up studies.

3.3. Tumor expression, normal selectivity, and risk landscape

Tumor expression was measured for 2,696 of 2,704 Core+Probable genes. The E component captured both expression magnitude and prevalence in TCGA-STAD primary tumors. The normal selectivity component N combined tumor versus GTEx stomach contrast, tumor versus mapped critical-normal contrast, and a pre-specified statistical rule. Under the GTEx stomach statistical selectivity rule, 757 genes met $\log_2 \mathrm{FC} \geq 1$ and BH-FDR < 0.05.

The off-tumor risk component R provided an opposing high-worse layer (Fig. 3). In the current conservative maximum-risk model, 1,825 Core+Probable genes had high critical off-tumor risk. This high count is expected because many cell-surface proteins are shared across normal epithelial, endothelial, hematopoietic, immune, or organ-specific compartments. E, N, and R therefore need to be interpreted together: tumor abundance without normal-context review can over-rank broadly expressed targets, while a high normal-risk flag does not automatically exclude gastric-lineage or clinically benchmarked antigens.

3.4. Integrated evidence across candidates

The primary frozen ranking was converted into coarse tiers only after stability analysis and manual curation (Fig. 4; Table 3). Tier 1 contains six unordered surface-target hypotheses: *ITGB4*, *CDH3*, *NECTIN2*, *CEACAM5*, *JAG1*, and *EPCAM*. Tier 2 contains twelve additional candidates: *MPZL1*, *ITGB5*, *BST2*, *IFNGR1*, *ALPG*, *DSC2*, *TNFRSF11A*, *ERBB2*, *CD9*, *LSR*, *CDH17*, and *TGFBR1*.

The Tier 1 set is intentionally not presented as a fine rank order. Its members combine high tumor expression/selectivity, protein/localization support, and accessible topology, but each carries interpretation caveats. For example, *ITGB4* has basal/stratified epithelial expression not fully captured by the organ-risk model, *NECTIN2* has a purity-adjusted myeloid/DC signal below the frozen high-TME threshold, *CEACAM5* and *EPCAM* retain normal epithelial expression concerns, and *JAG1* has dual epithelial/endothelial biology. These caveats are part of the prioritization output rather than post-hoc exclusions.

Tier 2 captures candidates with strong but less stable or more caveated evidence. *ERBB2*, an established HER2 benchmark, remains Tier 2 because its primary-scenario top-20 stability frequency (0.34) does not meet the pre-specified Tier 1 stability threshold of 0.40, although it is robustly recovered within the top 50. *CDH17* is likewise retained as Tier 2: its external clinical and biological relevance is noted, but the bulk ranking and stability criteria do not support Tier 1 placement in this pipeline.

3.5. TME specificity and compartment caveats

Because no processed gastric single-cell matrix was admitted into the primary score, compartment specificity was handled through flags rather than a numeric *SC* layer (Fig. 5; Supplementary File 2). The TME fallback identified known immune, endothelial, and stromal markers as high-risk cases for epithelial tumor-target interpretation. *PECAM1* remained high in the bulk ranking but was assigned to Watchlist by the pre-specified TME rule after curation confirmed vascular/endothelial rather than tumor-epithelial staining. *LRRC15* was also assigned to Watchlist because curation supported a CAF/stromal interpretation rather than a primary malignant-epithelial target claim.

The same compartment discipline applies to immune genes and thin-evidence high-rank artifacts. HLA and KIR genes with missing primary expression/protein/risk components were Watchlist by the missing-data rule, not by subjective demotion. *HLA-DPB1*, *IL2RG*, and related immune-context candidates are retained as biologically relevant but not nominated as epithelial tumor-cell surface targets.

The candidate-level TISCH2 cross-check is deliberately context-only. In `STAD_GSE134520`, all 18 Tier 1/2 genes were present: 8 were malignant-class supported, 1 was mixed, 7 had low malignant signal, and 2 were non-malignant dominant. `STAD_GSE167297` provided epithelial/TME context only because no malignant-cell class was present. These results are candidate annotations, not score components.

These single-cell calls are not symmetric evidence for or against tier placement. `STAD_GSE134520` contains only 880 malignant-class cells and was designed around premalignant lesions and early gastric cancer. Low or non-malignant calls for *CDH3*, *ALPG*, *CEACAM5*, and *EPCAM* therefore lower confidence in tumor-cell attribution and make tumor-cell membrane confirmation mandatory before experimental prioritization, but they do not override the frozen multi-layer score or tiers.

NECTIN2 and *JAG1* illustrate the same rule in the opposite direction: supportive annotations can reduce concern but do not eliminate compartment caveats. Both remain Tier 1 under frozen criteria, but *NECTIN2* remains closest to the TME threshold and *JAG1* remains exposed to endothelial/angiocrine interpretation.

3.6. Stability and control diagnostics

Stability analysis supported coarse candidate interpretation but not fine within-tier ordering (Fig. 6a-b). Weight perturbations were generally stable: 231 of 250 perturbations passed both pre-specified gates, with minimum and median Spearman correlations of 0.910703 and 0.983898 relative to the final ranking. However, leave-one-layer-out analyses showed important layer dependence, especially for T, R, and E, and post-scoring rank resolution found only 1 of 20 baseline top20 genes with a 95% rank interval contained within the top 40. These results justify coarse tiers rather than fine rank claims.

Control behavior was used as a diagnostic rather than for weight tuning (Fig. 7; Table 4). Positive-control top50 recovery remained imperfect at 4 of 8 (*ERBB2*, *EPCAM*, *MET*, and *CEACAM5*). The aggregate pre-specified top50 gate therefore remains reported as failed. The cause review was added after the aggregate miss and is reported as an explanatory diagnostic, not as a replacement success criterion: *CLDN18* and *FGFR2* are isoform unresolved at gene level, *MSLN* retains a topology/shedding penalty despite high protein support and corrected GPI evidence, and *TACSTD2* has intermediate surface confidence and weaker protein support. Negative intracellular controls were excluded from Core/Probable; *PTPRC* was cleanly low-ranked for epithelial targeting; *PECAM1* remained the documented vascular/TME exception and was handled by Watchlist tiering.

3.7. Candidate interpretation

The candidate panel summarizes the final applied output of the pipeline (Fig. 8; Table 3; Supplementary File 2). The Tier 1 set should be read as a compact, experimentally testable hypothesis set rather than as a list of proven therapeutic targets. Candidate behavior is most useful where it reveals the framework's boundaries: *ERBB2* remaining Tier 2 shows that benchmark status did not override frozen stability rules, and Watchlist placement for *PECAM1* and *LRRC15* shows how bulk surfaceome ranking can over-rank endothelial or stromal signals when cell-resolved data are unavailable.

Tier 1/2 candidates therefore function as a prioritized experimental queue with explicit caveats. Their value in this manuscript is not stand-alone translational validity, but the fact that each nomination carries the evidence layers, missingness, compartment warnings, normal-risk context, and stability limits needed to decide what validation would be required next.

4. Discussion

4.1. What the pipeline adds

This study presents SurfPrior-GC, a reproducible public-data workflow for prioritizing gastric cancer surface-target hypotheses while preserving the uncertainty that usually gets compressed into a single ranked list. Its main positive methods contribution is the GPI-routing audit: 54 confirmed GPI-anchor genes were absent from the ranked universe under topology-first routing, and universe-stage correction changed both candidate-space membership and benchmark ranks before final scoring. The broader pipeline integrates surfaceome membership, tumor expression, tumor-normal selectivity, normal-tissue risk, protein/localization evidence, and topology/accessibility, while keeping single-cell specificity unavailable rather than imputed. Its contribution is not a claim that any individual candidate is ready for translation, but a transparent method for combining heterogeneous evidence layers and showing where that evidence is strong, weak, or confounded. This positioning is deliberately narrower than pan-cancer target-discovery frameworks, tumor-specificity models, general multi-omics prioritization platforms, and recent gastric target-prioritization or proteogenomic resources (Li et al., 2022; Razzaghdoust et al., 2021; Kathad et al., 2024; Deng et al., 2024; Moek et al., 2017; Li et al., 2024; Xu et al., 2025; Chang et al., 2026).

The TCSA and Wang comparisons address different external-reference questions. TCSA provides a pan-cancer surfaceome-score baseline: the moderate rank correlation and low top20 identity show that the workflow is aligned with established surfaceome evidence but not reducible to it. Wang 2026 is best interpreted as a gastric-specific consistency check. Other contemporary gastric proteogenomic work provides therapeutic-vulnerability context but is not treated here as a candidate label (Chang et al., 2026). The overlap with Wang exceeds a simple random draw but not a matched null, which matters because both resources enrich for highly expressed and well-annotated surface proteins. Taken together, the external checks support plausibility and benchmark consistency, while CD9 and LSR remain divergent nominations, glyco-outlier corroboration is selective, and the curated Wang Figure 7H single-cell panel does not directly plot most of the specific nominees.

Recent gastric/GEJ/GSRCC ADC studies show why the present work should not be framed as first target discovery. CD66c and PVR/CD155 have already been prioritized and tested in subtype- or site-focused ADC workflows, and CDH17-targeted ADC work illustrates continued target-specific development in gastrointestinal cancers (Zhang et al., 2025; Zhao et al., 2026; Wang et al., 2025). The value of the present framework is complementary: it makes the broad public-data prioritization process, evidence routing, stability limits, and candidate caveats auditable before any wet-lab or modality-specific validation is claimed.

The workflow also makes uncertainty visible. ERBB2 remaining Tier 2 rather than being forced into Tier 1 shows that benchmark recovery was not used to override frozen stability rules. PECAM1 and LRRC15 moving to Watchlist shows that bulk RNA surfaceome ranking can produce compartment-derived false positives when cell-resolved data are unavailable. CLDN18 and FGFR2 remaining isoform unresolved shows that gene-level expression is insufficient for isoform-specific claims. These are not failures to be hidden; they are the evidence boundaries that make the prioritization interpretable.

4.2. Transferable GPI-anchor lesson

The GPI-anchor audit is the most transferable concrete methods artifact from this work. Confirmed GPI-anchored proteins are biologically membrane-associated and can be relevant to antibody-accessible targeting, and prior surfaceome work already recognized this class explicitly (Bausch-Fluck et al., 2018). The narrower lesson is that a pipeline can under-credit GPI-anchored candidates when list overlap or classical transmembrane topology is weighted above confirmed lipidation evidence during universe construction.

The correction is therefore a routing-design precaution for workflows that build candidate universes from heterogeneous surfaceome lists and topology fields. It does not generalize to resources that already handle GPI evidence appropriately, such as SURFY, and it does not imply that GPI anchoring alone makes a good target. The discussion-level consequence is compact: evidence routing changed which 54 confirmed GPI-anchor genes the framework was allowed to evaluate before scoring began.

4.3. Candidate tiers as method diagnostics

The candidate tiers are useful because they show how the framework behaves under pressure. Established biology does not automatically override frozen criteria: ERBB2 remains Tier 2 because it misses the Tier 1 stability threshold. High bulk expression does not automatically imply malignant epithelial targeting: PECAM1 and LRRC15 are retained as Watchlist examples of endothelial and stromal signal. External concordance does not erase normal-tissue or compartment caveats: CEACAM5, EPCAM, NECTIN2, ITGB4, JAG1, and CDH3 still require candidate-specific validation before any translational claim.

This candidate behavior is the point of a methods-forward paper. The output is an experimental queue with evidence boundaries, not a ranked claim that the top genes are intrinsically superior therapeutic targets. Cases such as ERBB2, PECAM1, LRRC15, CLDN18, FGFR2, and the GPI-anchored candidates show where the framework refuses to simplify away benchmark expectations, compartment risk, isoform ambiguity, or routing artifacts.

The Watchlist is therefore not a discard bin. It is a visible record of the cases that a score-only view could over-promote, including immune HLA/KIR genes with thin evidence, endothelial PECAM1, CAF/stromal LRRC15, immune IL2RG/HLA-DPB1/BTN3A3, and ubiquitous or toxicity-prone CD47.

4.4. Limitations

Several limitations are central to interpretation. First, the Tier 1/2 set is externally consistent, but its Wang overlap exceeds a simple random-draw null and not the matched null that controls for surfaceome confidence, expression, protein evidence, topology, component completeness, and accessibility. This is a boundary made visible by the framework, not a secondary apology for the result.

Second, the expression layers are based on bulk RNA rather than isolated malignant epithelial cells. Bulk signal can reflect tumor cells, stromal cells, immune cells, endothelial cells, or mixtures of these compartments (Yoshihara et al., 2013; Han et al., 2023; Zhang et al., 2019; Jeong et al., 2021). Bulk TME correlations, ESTIMATE-adjusted partial correlations, and the limited TISCH2 candidate-level check provide conservative review flags, but they do not replace single-cell or spatial validation.

Third, tumor-normal selectivity is constrained by source and sample-size limitations. TCGA-STAD primary tumors and GTEx stomach normals were processed through the Xena/Toil recompute, but TCGA/GTEx source effects remain explicit. TCGA adjacent normal samples provide a source-matched sensitivity context but are lower-powered than GTEx stomach. The risk model is intentionally conservative, yet some normal compartments such as skin/basal epithelium are incompletely represented and must be revisited in candidate-specific review.

Fourth, HPA bulk IHC provides useful protein and localization evidence but lacks antibody-level and patient-level membrane detail in the downloadable cancer file (Uhlen et al., 2015, 2017; Thul et al., 2017). A candidate with HPA tumor staining and subcellular membrane support still requires antibody-specific review, tumor-cell membrane quantification, and normal-tissue cross-reactivity assessment before experimental prioritization.

Fifth, the primary score did not admit a processed gastric scRNA dataset. SC is therefore unavailable rather than imputed. The limited TISCH2 analysis annotates compartment caveats for the 18 Tier 1/2 candidates, but it does not solve cell-of-origin for the full ranked universe, provide patient-level malignant-cell prevalence for every candidate, or alter scores or tiers (Han et al., 2023; Zhang et al., 2019; Jeong et al., 2021). Only `STAD_GSE134520` contributes a TISCH2 malignant-cell class, and `STAD_GSE167297` is context-only.

Sixth, gene-level expression cannot resolve isoform-specific targets. CLDN18.2 and FGFR2b remain explicitly `isoform_unresolved`; their positions in the ranking cannot be used as evidence that the pipeline resolves isoform-specific abundance. Similarly, topology/accessibility annotations do not establish antigen density, epitope exposure in native tumor tissue, internalization, shedding behavior, or therapeutic index.

Finally, this is a computational, hypothesis-generating study without wet-lab validation. Wang 2026 consistency checks and HPA/UniProt support strengthen candidate context, but they do not establish clinical efficacy or safety for any nominated target.

Sex-stratified and gender-related analyses were outside the scope of the current analysis. These dimensions should be assessed where relevant in future candidate-specific validation and translational study design.

4.5. Experimental follow-up

The highest-priority follow-up is experimental confirmation of tumor-cell surface abundance. Flow cytometry, cell-surface capture proteomics, or orthogonal surface-enrichment methods would directly test the surface-accessibility assumptions that public RNA, HPA, and topology annotations cannot resolve alone (Bausch-Fluck et al., 2015; Uhlen et al., 2015; Thul et al., 2017). Tumor and normal tissue IHC should be repeated with validated antibodies and scored for membrane pattern, tumor-cell specificity, and normal compartment expression.

Compartment resolution is especially important for the most exposed candidates. NECTIN2 should be evaluated in cell-resolved data to distinguish tumor-epithelial expression from dendritic/myeloid checkpoint-ligand biology. JAG1 requires separation of tumor epithelial and endothelial/vascular expression. PECAM1 and LRRC15 provide useful negative examples for epithelial targeting but may be relevant to vascular or stromal strategies if framed as different modalities.

Modality-specific assays should follow confirmation of tumor-cell surface abundance. ADC-relevant candidates require internalization and payload-sensitivity testing. CAR or bispecific candidates require antigen-density and normal cross-reactivity assessment. GPI-anchored candidates require attention to shedding, soluble isoforms, and surface-retention behavior. The output of this pipeline is therefore a prioritized experimental queue, not a substitute for target validation.

5. Conclusions

This study provides SurfPrior-GC, a reproducible framework for uncertainty-aware gastric cancer surface-target prioritization from public evidence. Its concrete transferable contribution is the GPI audit: confirmed lipidation evidence should be handled at universe construction when defining surfaceome candidates, because incomplete routing excluded 54 confirmed GPI-anchor genes and changed benchmark-relevant ranks.

The resulting Tier 1 and Tier 2 hypotheses are a coarse experimental queue, not a target list alone. They preserve a transparent record of what heterogeneous bulk RNA, protein/localization, topology, normal-risk, surfaceome, and benchmark data can and cannot establish.

The matched-null result defines the scope of the evidence: gastric-atlas agreement is not enriched beyond a surfaceome-evidence-matched null, and no wet-lab validation is provided here. The nominated genes should therefore be treated as experimentally testable hypotheses with explicit validation requirements, not as established therapeutic targets.

6. Glossary

ADC. Antibody-drug conjugate.

CSPA. Cell Surface Protein Atlas.

GPI. Glycosylphosphatidylinositol, a lipid anchor that can attach proteins to the cell surface.

GTEx. Genotype-Tissue Expression project.

HPA. Human Protein Atlas.

IHC. Immunohistochemistry.

SC. Single-cell specificity score component; unavailable and not imputed in the primary score.

SURFY. Published in silico human surfaceome resource.

TCSA. The Cancer Surfaceome Atlas.

TISCH2. Tumor Immune Single-cell Hub 2.

TME. Tumor microenvironment.

7. Figure captions

Figure 1. SurfPrior-GC reproducible public-data workflow for gastric cancer surface-target prioritization. The workflow begins with frozen public inputs, identifier normalization, and construction of a Core+ Probable surfaceome universe, then computes tumor expression (E), normal selectivity (N), high-worse off-tumor risk (R), HPA protein/localization evidence (P), topology/accessibility (T), TME flags, stability checks, and coarse tiering. The figure reports stage-level counts and separates quantitative evidence layers from review flags and downstream hypothesis generation.

Figure 2. SurfPrior-GC surfaceome evidence landscape and GPI-anchor correction. Source overlap and confidence summaries show how curated surfaceome resources, UniProt topology/GPI evidence, HPA localization, GO annotations, and SURFY support contribute to the 2,704-gene Core+Probable universe. Confirmed UniProt lipidation GPI-anchor evidence is credited as a source-wide non-experimental strong anchor during universe construction. Low $Surf$ values in the final ranking indicate lower relative confidence within the admitted universe, not absence from the surfaceome.

Figure 3. Tumor-normal selectivity and off-tumor risk landscape. Candidate genes are positioned by TCGA-STAD tumor expression, stomach and critical-normal selectivity, and conservative organ-risk context. R is a high-worse component and is subtracted in the integrated score. The display illustrates why tumor abundance, normal selectivity, and off-tumor risk must be interpreted jointly rather than as independent target-readiness claims.

Figure 4. Multi-layer evidence heatmap for the top 30 ranked candidates. The heatmap summarizes $Surf$, E, N, R, P, T, missingness, TME flags, topology/accessibility, and coarse tiers for the top-ranked candidates from the final frozen ranking. Tiers are unordered within Tier 1, Tier 2, and Watchlist. Watchlist entries document immune, stromal, endothelial, ubiquitous, or thin-evidence cases that would be overinterpreted by a score-only view.

Figure 5. Tumor microenvironment specificity fallback. Bulk TCGA-STAD marker-module and ESTIMATE-adjusted partial-correlation outputs flag likely CAF/fibroblast, endothelial, myeloid/macrophage, T-cell, and B/plasma compartment signals among top candidates. These outputs are flags only and do not replace cell-resolved single-cell or spatial validation. SC remains `not_available` and is not imputed in the primary score.

Figure 6. Rank stability and weighting sensitivity. (a) Rank-stability heatmap from weight perturbation, leave-one-layer-out, missing-data, risk-form, organ-weight, and post-scoring resolution analyses. (b) Weighting-sensitivity bumpchart comparing the primary pre-specified ranking with safety-prioritized, ADC-focused, novelty-focused, protein-evidence-focused, and robust-aggregate contexts. Stability supports coarse tiers and explicit caveats, but not fine within-tier ordering.

Figure 7. Benchmark-control behavior. Positive, secondary, negative, and TME/off-tumor controls are shown as diagnostics of pipeline behavior rather than as tuning targets. Positive-control top50 recovery is reported as imperfect, isoform-specific benchmarks remain unresolved where expression is gene-level, intracellular negative controls are excluded from Core+Probable, and *PECAM1* is retained as the documented vascular/TME exception handled by Watchlist tiering.

Figure 8. Tier 1 candidate panel. The six unordered Tier 1 hypotheses are *ITGB4*, *CDH3*, *NECTIN2*, *CEACAM5*, *JAG1*, and *EPCAM*. The panel summarizes component evidence, Wang 2026 concordance, and candidate-specific caveats. These nominations are experimentally testable

hypotheses and do not imply clinical efficacy, clinical safety, antigen density, internalization, or wet-lab validation.

8. Table captions

Table 1. Public datasets, versions, uses, and limitations. Primary and secondary sources used for the workflow, including Xena/Toil TCGA-STAD/GTEx RNA, GDC and cBioPortal metadata, HPA IHC/localization/RNA files, UniProt reviewed-human topology/GPI features, curated surfaceome resources, TISCH2 candidate-level scRNA context, and optional resources not admitted into the numeric primary score. The table records source versions, checksum manifests where applicable, and interpretation limits.

Table 2. Score components and primary weights. Definitions, directions, weights, primary sources, normalization conventions, and principal limitations for $Surf$, E, N, R, P, SC, and T. R is high-worse and subtracted; SC remains unavailable with zero weight; missing components are handled by exclude-and-renormalize during scoring and by tier restrictions during interpretation.

Table 3. Tier 1 and Tier 2 candidates. Coarse unordered Tier 1 and Tier 2 nominations from the final frozen ranking, stability outputs, and manual curation. Columns report rank, score, prevalence flag, stability frequency, component values, Wang 2026 class/glyco-outlier context, and the principal caveat. The table is a hypothesis list, not a clinical target list.

Table 4. Benchmark-control diagnostics. Positive controls, secondary benchmarks, intracellular negative controls, and TME/off-tumor controls used to assess whether the pipeline behaves plausibly. Control behavior is reported honestly, including the failed aggregate positive-control top50 gate and the cause-level interpretation of misses. Controls were not used to retune weights or rescue individual candidates.

9. Supplementary material

Supplementary material is supplied separately for Editorial Manager. Supplementary File 1 contains the table manifest for Supplementary Tables S1-S24, including titles, repository paths, availability status, and archived-data-package coverage. Supplementary File 2 contains candidate flags and interpretation guardrails for the top 30 ranked candidates, including TME risk, normal-risk interpretation, accessibility class, HPA status, discordance flags, missing primary components, isoform-resolution status, and candidate-specific caveats.

10. Data and code availability

All inputs used in the current analysis are public resources. Downloaded or captured raw/source files are listed in the release manifest, dataset registry, download manifest, and provenance log, with checksums where files are stored locally. Mutable API captures, including cBioPortal clinical/GISTIC JSON files, are treated as frozen archived inputs for release reproduction; live re-downloads are best-effort transparency checks rather than the default reproducibility claim.

Analysis code is available at https://github.com/vicenzoscavino1999/surfaceome-gastric-cancer under final code release tag `v0.1.3`. The frozen reproducibility data package for release candidate `v0.1.0-rc4` is archived on Zenodo at https://zenodo.org/records/20498705 with DOI `10.5281/zenodo.20498705`. The active frozen ranking table has SHA256 `95040edef1b2a50c9ab1d61042856485bbfcac98e9f51d4cb78cbe05c46e9631`, with file-level metadata in `results/rankings/ranking_v2_frozen.metadata.yaml`.

The repository includes smoke tests, unit tests, manuscript checks, Snakemake dry-run checks, GitHub Actions small CI, and manual full-data release-audit hooks documented in the README and reproducibility guide. No new patient samples, wet-lab data, or restricted-access clinical data were generated for this study.

11. Declarations

Ethics approval and consent to participate. Not applicable. This study uses publicly available, de-identified, aggregate, or open-access research data and does not generate new human-subject samples.

Consent for publication. Not applicable.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process. During the preparation of this work, the author used OpenAI Codex to support manuscript organization, language editing, and repository-oriented consistency checks. The author reviewed, revised, verified, and approved the final manuscript and accepts full responsibility for its content.

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Review figures and main tables

The following pages embed the main display items for the review PDF. The same figures and editable table files are also supplied separately in the Editorial Manager package.

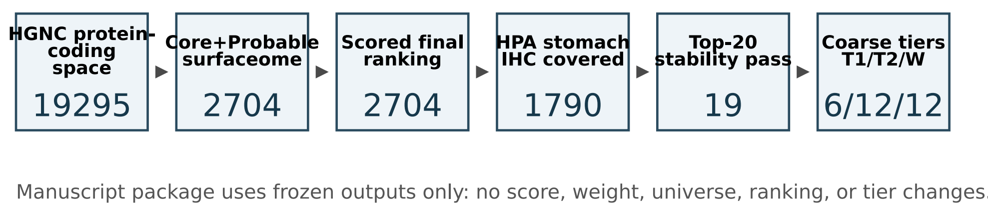


Figure 1: SurfPrior-GC reproducible public-data workflow for gastric cancer surface-target prioritization. The workflow begins with frozen public inputs, identifier normalization, and construction of a Core+Probable surfaceome universe, then computes tumor expression (E), normal selectivity (N), high-worse off-tumor risk (R), HPA protein/localization evidence (P), topology/accessibility (T), TME flags, stability checks, and coarse tiering. The figure reports stage-level counts and separates quantitative evidence layers from review flags and downstream hypothesis generation.

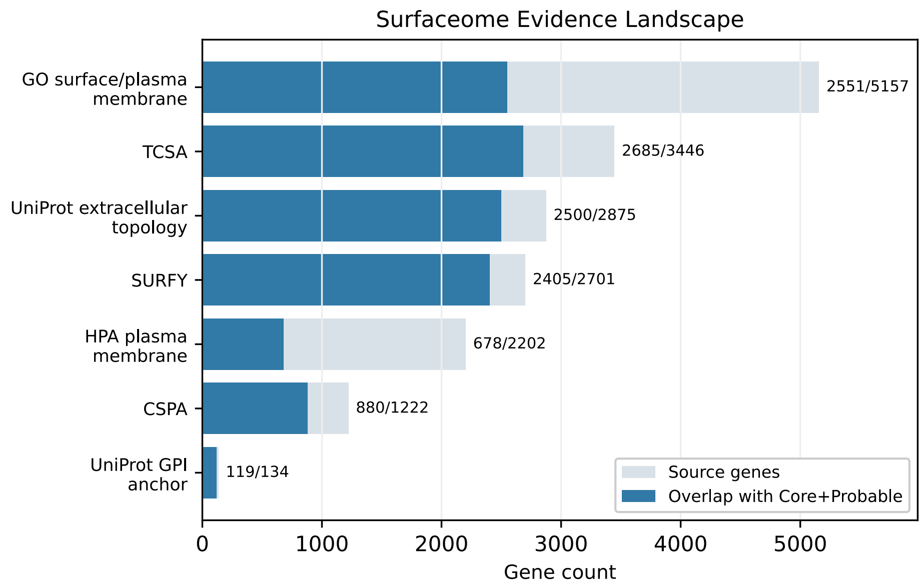


Figure 2: SurfPrior-GC surfaceome evidence landscape and GPI-anchor correction. Source overlap and confidence summaries show how curated surfaceome resources, UniProt topology/GPI evidence, HPA localization, GO annotations, and SURFY support contribute to the 2,704-gene Core+Probable universe. Confirmed UniProt lipidation GPI-anchor evidence is credited as a source-wide non-experimental strong anchor during universe construction. Low $Surf$ values in the final ranking indicate lower relative confidence within the admitted universe, not absence from the surfaceome.

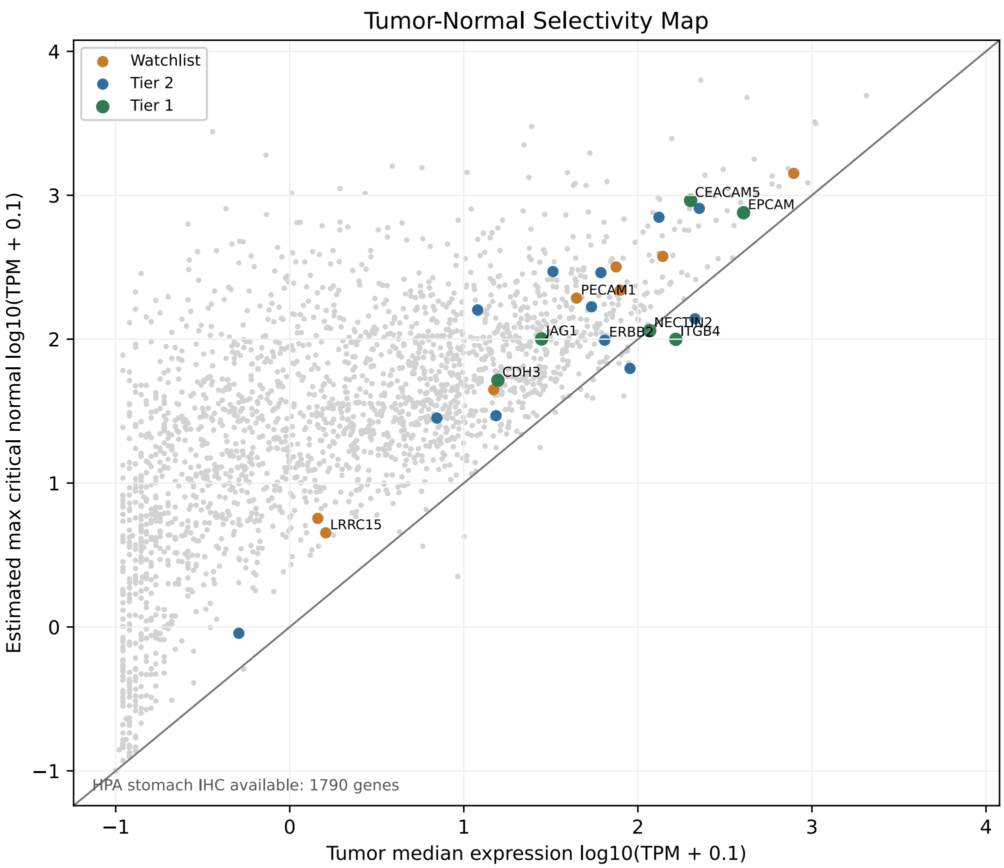


Figure 3: Tumor-normal selectivity and off-tumor risk landscape. Candidate genes are positioned by TCGA-STAD tumor expression, stomach and critical-normal selectivity, and conservative organ-risk context. R is a high-worse component and is subtracted in the integrated score. The display illustrates why tumor abundance, normal selectivity, and off-tumor risk must be interpreted jointly rather than as independent target-readiness claims.

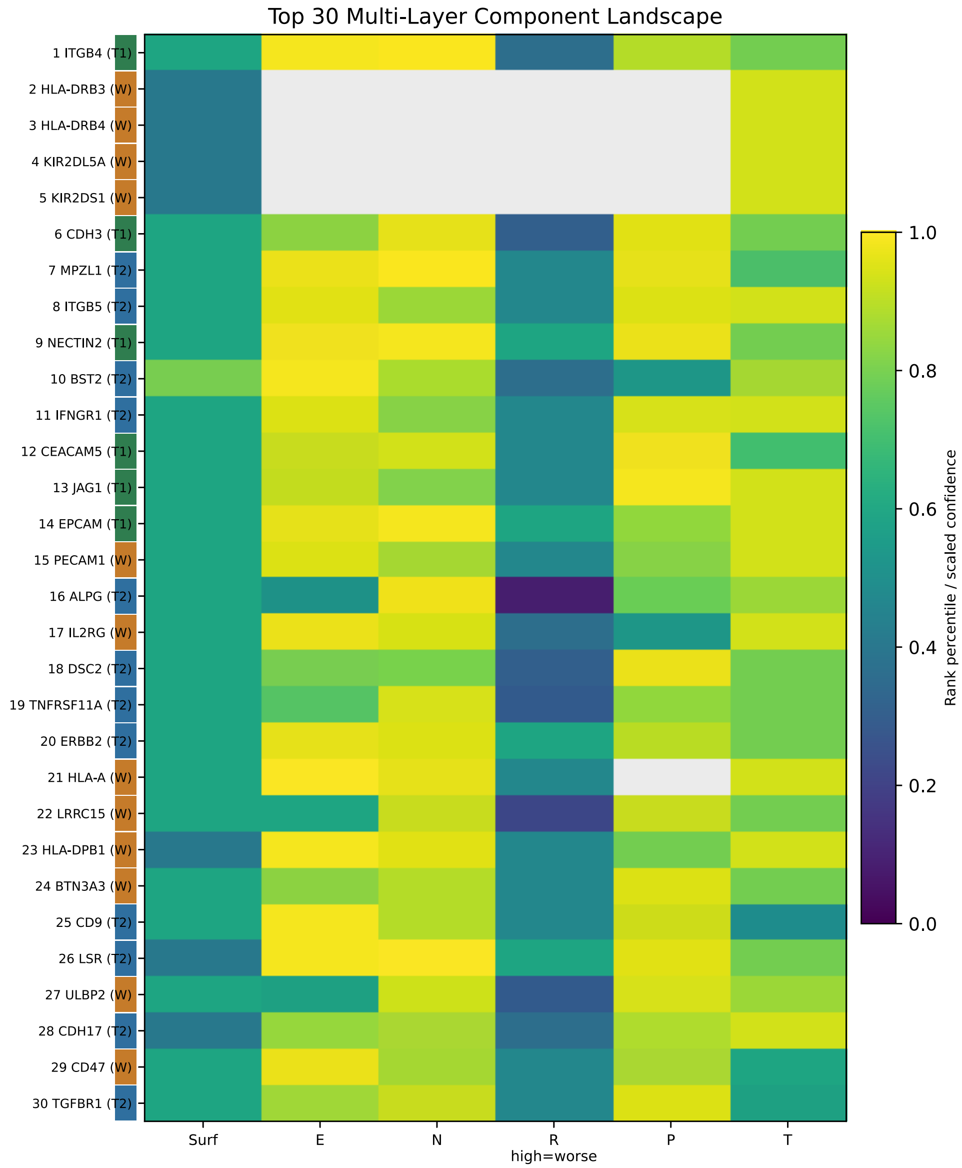


Figure 4: Multi-layer evidence heatmap for the top 30 ranked candidates. The heatmap summarizes $Surf$, E, N, R, P, T, missingness, TME flags, topology/accessibility, and coarse tiers for the top-ranked candidates from the final frozen ranking. Tiers are unordered within Tier 1, Tier 2, and Watchlist. Watchlist entries document immune, stromal, endothelial, ubiquitous, or thin-evidence cases that would be overinterpreted by a score-only view.

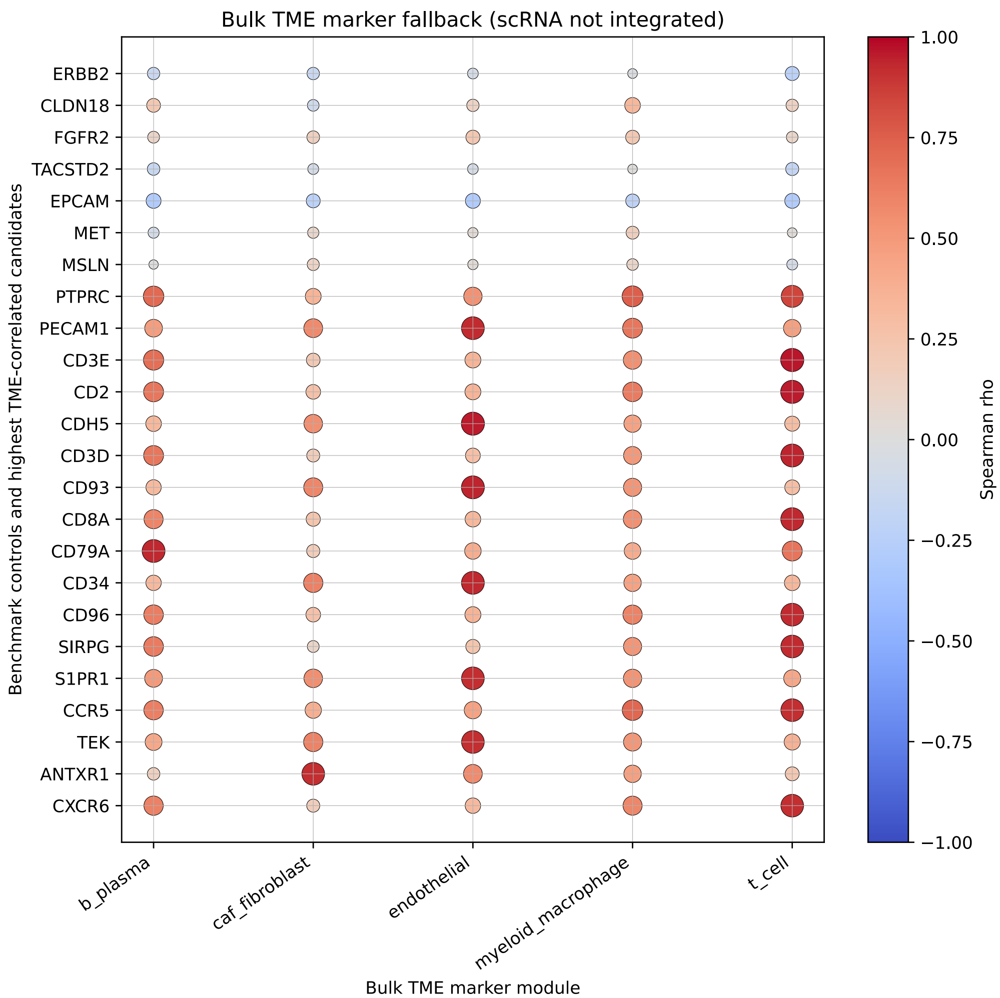
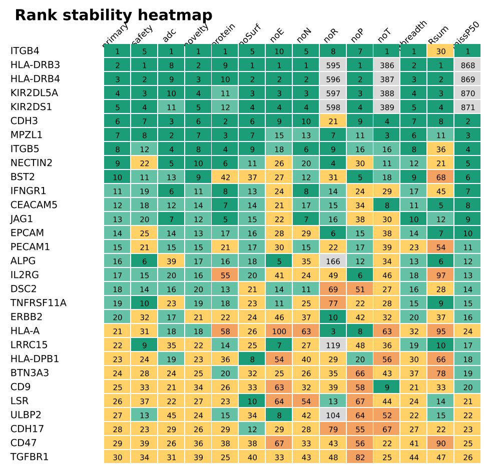


Figure 5: Tumor microenvironment specificity fallback. Bulk TCGA-STAD marker-module and ESTIMATE-adjusted partial-correlation outputs flag likely CAF/fibroblast, endothelial, myeloid/macrophage, T-cell, and B/plasma compartment signals among top candidates. These outputs are flags only and do not replace cell-resolved single-cell or spatial validation. *SC* remains `not_available` and is not imputed in the primary score.



a.

b.

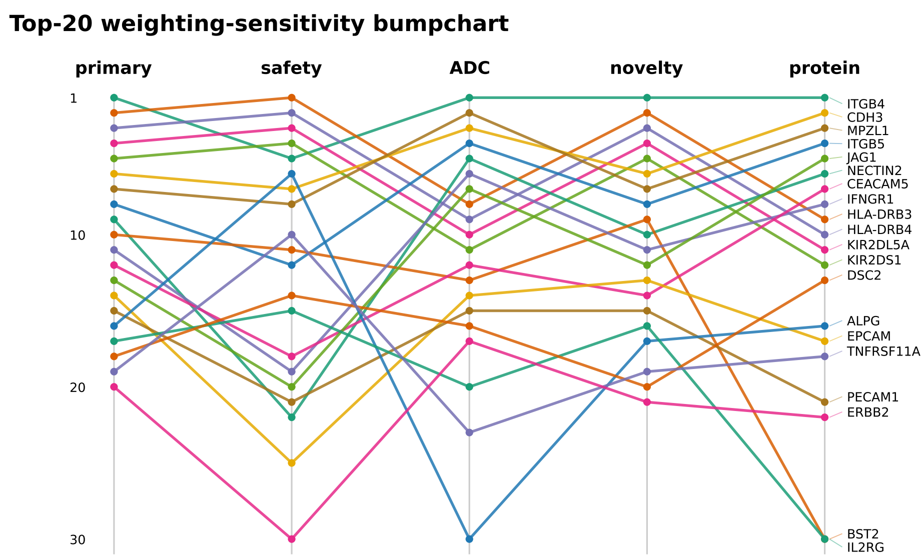


Figure 6: Rank stability and weighting sensitivity. (a) Rank-stability heatmap from weight perturbation, leave-one-layer-out, missing-data, risk-form, organ-weight, and post-scoring resolution analyses. (b) Weighting-sensitivity bumpchart comparing the primary pre-specified ranking with safety-prioritized, ADC-focused, novelty-focused, protein-evidence-focused, and robust-aggregate contexts. Stability supports coarse tiers and explicit caveats, but not fine within-tier ordering.

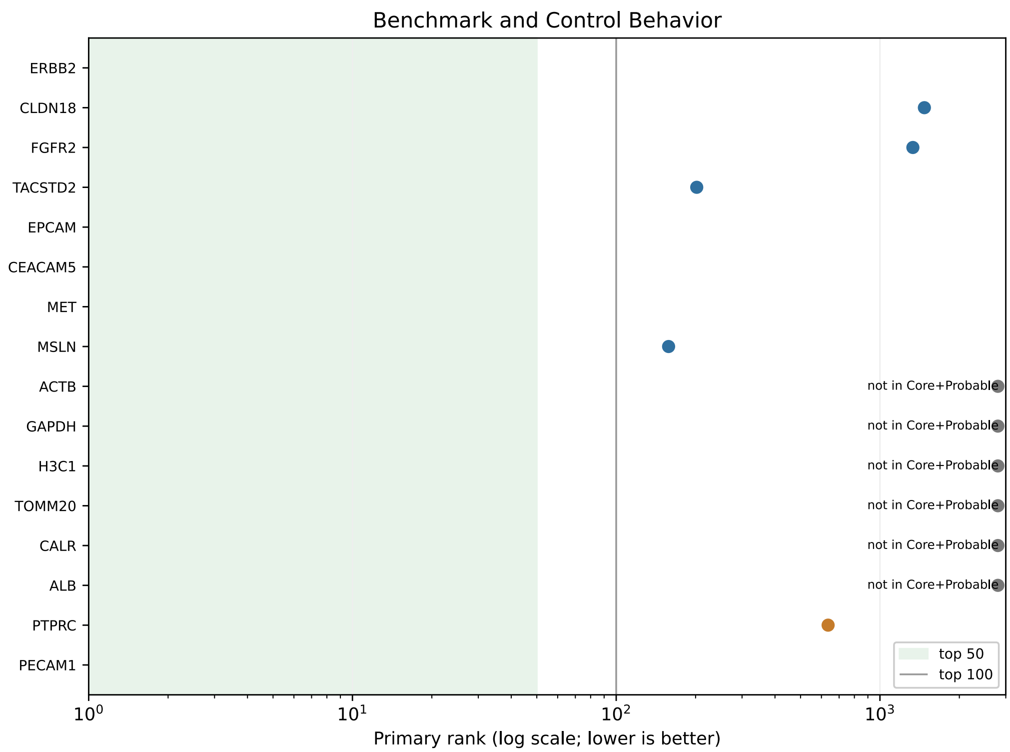


Figure 7: Benchmark-control behavior. Positive, secondary, negative, and TME/off-tumor controls are shown as diagnostics of pipeline behavior rather than as tuning targets. Positive-control top50 recovery is reported as imperfect, isoform-specific benchmarks remain unresolved where expression is gene-level, intracellular negative controls are excluded from Core+Probable, and *PECAM1* is retained as the documented vascular/TME exception handled by Watchlist tiering.

Tier 1 Candidate Summary (unordered within tier)

Gene	Rank	Score	Stability freq	Wang class	Main caveat
ITGB4	1	0.626035	1.000	Membrane	CAVEAT: basal/stratified epithelial (skin keratinocyte/hemidesmosome)...
CDH3	6	0.608164	1.000	Membrane	established TAA; confirm membranous tumor IHC in pass 2
NECTIN2	9	0.588737	1.000	Membrane	CAVEAT: purity-adjusted myeloid/DC partial 0.32 p<0.001 + LAMP3+ DC...
CEACAM5	12	0.584828	1.000	ADC/Antibody	apical/normal GI epithelial expression; specificity review
JAG1	13	0.584280	1.000	Membrane	CAVEAT: dual tumor-epithelial + endothelial/vascular (angiocrine)...
EPCAM	14	0.580488	1.000	ADC/Antibody	high normal epithelial expression (preregistered normal-expression...

Curation informs tier only; no score/rank/tier recalibration.

Figure 8: Tier 1 candidate panel. The six unordered Tier 1 hypotheses are *ITGB4*, *CDH3*, *NECTIN2*, *CEACAM5*, *JAG1*, and *EPCAM*. The panel summarizes component evidence, Wang 2026 concordance, and candidate-specific caveats. These nominations are experimentally testable hypotheses and do not imply clinical efficacy, clinical safety, antigen density, internalization, or wet-lab validation.

Table 1: Public datasets, versions, uses, and limitations. Primary and secondary sources used for the workflow, including Xena/Toil TCGA-STAD/GTEx RNA, GDC and cBioPortal metadata, HPA IHC/localization/RNA files, UniProt reviewed-human topology/GPI features, curated surfaceome resources, TISCH2 candidate-level scRNA context, and optional resources not admitted into the numeric primary score. The table records source versions, checksum manifests where applicable, and interpretation limits.

source id	source name	primary use	analysis stage	version or release	status	checksum manifest	limitation or note
xena_toil_tcga_gtex	UCSC Xena / Toil RNA-seq recompute	Primary TCGA-STAD vs GTEx stomach RNA matrix	source acquisition download; inventory metadata complete	Toil recompute; matrix Last-Modified Fri, 09 Apr 2021 20:01:53 GMT	metadata_verified_no_matrix_downloaded	data/checksums/xena_toil_sha256.tsv	Expression unit to register as log2(TPM+0.001); phenotype counted without downloading matrix.
gdc_tcga_stad	GDC TCGA-STAD API	TCGA clinical, biospecimen, STAR-count RNA-seq sensitivity	inventory metadata complete; source acquisition selected downloads	GDC API live query	metadata_verified_no_raw_downloaded	data/checksums/gdc_tcga_stad_sha256.tsv	GDC STAR counts are secondary sensitivity, not the primary TCGA-GTEx matrix.
cbioportal_stad_tcga	cBioPortal TCGA-STAD clinical	TCGA subtype and legacy clinical field availability	inventory metadata complete	stad_tcga and stad_tcga_pan_can_atlas_2018 API studies	metadata_verified	data/checksums/cbioportal_sha256.tsv	Used for TCGA molecular subtype counts and legacy histology fields.
hpa_downloads	Human Protein Atlas downloadable TSV	Normal IHC, cancer IHC, subcellular location, tissue RNA cross-checks	inventory metadata complete; source acquisition downloads	HPA 25.1; Ensembl 109	metadata_verified_no_raw_saved	data/checksums/hpa_sha256.tsv	HPA final5.1 replaces older plan filenames normal_tissue.tsv/pathology.tsv with current normal_ihc_data.tsv/cancer_data.tsv.
uniprot_reviewed_human	UniProtKB reviewed human	Protein identifiers, topology, TM, extracellular domains, signal peptides	inventory metadata complete; surfaceome universe ID mapping/topology	2026_01	metadata_verified	data/checksums/uniprot_sha256.tsv	Universe-specific reviewed coverage waits for HGNC/Ensembl/UniProt mapping.
depmap	DepMap Portal Context Explorer	Gastric/GEJ cell-line expression and CRISPR dependency evidence	inventory metadata complete; inventory1 optional annotation	DepMap Portal current release; 26Q1 label observed on portal	metadata_verified	data/checksums/depmap_sha256.tsv	104 Esophagus/Stomach models; EGC subset is the closest adenocarcinoma set.
pdc_cptac_gastric	NCI Proteomic Data Commons gastric/STAD studies	Incremental proteomics, phosphoproteomics, glycoproteomics, and related omics	inventory metadata complete; optional after HPA/scRNA/external RNA	PDC GraphQL live query	metadata_verified_no_raw_downloaded	data/checksums/pdc_cptac_sha256.tsv	Candidate gene protein coverage remains TBD until Protein Assembly reports are downloaded and mapped.
scrna_gastric_candidates	TISCH2/GEO gastric cancer scRNA candidates	Malignant epithelial vs TME specificity layer	inventory candidate inventory; TME specificity fallback quality gate	GEO/TISCH2 current at access date	candidate_sources_identified_not_downloaded	data/checksums/scrna_sha256.tsv	Do not include in main score until annotation quality and processed matrix availability pass gate.
external_bulk_rna_cohorts	GEO/ACRG external gastric expression cohorts	External RNA validation cohort	inventory candidate inventory; tumor expression-7 optional validation	GEO current at access date	candidate_sources_identified_not_downloaded	data/checksums/external_bulk_rna_sha256.tsv	Layer is incremental; avoid delaying primary if batch/platform harmonization becomes expensive.
structure_sources	RCSB PDB and AlphaFold DB	Candidate-level ECD/structure annotations	inventory0 candidate-level optional layer	Current APIs at candidate query date	source_declared_not_assessed	data/checksums/structures_sha256.tsv	Not needed before scoring; used for candidate cards and modality discussion.

Table 1: Public datasets, versions, uses, and limitations. Primary and secondary sources used for the workflow, including Xena/Toil TCGA-STAD/GTEx RNA, GDC and cBioPortal metadata, HPA IHC/localization/RNA files, UniProt reviewed-human topology/GPI features, curated surfaceome resources, TISCH2 candidate-level scRNA context, and optional resources not admitted into the numeric primary score. The table records source versions, checksum manifests where applicable, and interpretation limits. (continued)

source id	**source name**	**primary use**	**analysis stage**	**version or release**	**status**	**checksum manifest**	**limitation or note**
clinical_druggability_sources	Open Targets, ChEMBL, DGIdb, ClinicalTrials.gov	Clinical/druggability annotations	inventory2 candidate-level annotation	Current APIs at candidate query date	source_declared_not_assessed	data/checksums/clinical_druggability_sha256.tsv	Used for candidate cards and tiering context, not to inflate discovery score.

Table 2: Score components and primary weights. Definitions, directions, weights, primary sources, normalization conventions, and principal limitations for $Surf$, E, N, R, P, SC, and T. R is high-worse and subtracted; SC remains unavailable with zero weight; missing components are handled by exclude-and-renormalize during scoring and by tier restrictions during interpretation.

component	**definition**	**primary weight**	**direction**	**primary source**	**normalization**	**main limitation**
Surf	surfaceome membership confidence; higher is better	0.150000	higher_better	surfaceome_universe.tsv	rank_percentile_or_fixed_scale	Membership confidence within admitted Core+Probable universe; not direct antigen density.
E	tumor expression and prevalence; higher is better	0.200000	higher_better	TCGA-STAD tumor_expression.tsv	rank_percentile_or_fixed_scale	Bulk RNA; no malignant epithelial isolation.
N	tumor-normal selectivity; higher is better	0.200000	higher_better	selectivity_scores.tsv	rank_percentile_or_fixed_scale	TCGA/GTEx source effects remain explicit.
R	on-target/off-tumor risk; higher is worse and is subtracted	0.200000	higher_worse_subtracted	off_tumor_risk.tsv	rank_percentile_or_fixed_scale	High-worse off-tumor risk is subtracted; risk form is sensitivity-tested.
P	protein/localization evidence; higher is better	0.150000	higher_better	protein_evidence.tsv	rank_percentile_or_fixed_scale	HPA bulk IHC lacks antibody-level and patient-level membrane fields.
SC	tumor-cell specificity versus TME; higher is better; optional in primary	0.000000	higher_better	single_cell_specificity.tsv (not_available in primary score)	rank_percentile_or_fixed_scale	No processed annotated gastric scRNA matrix admitted into the score; not counted in the primary score.
T	topology, extracellular accessibility, and isoform confidence; higher is better	0.100000	higher_better	topology_isoforms_ecd.tsv	rank_percentile_or_fixed_scale	Topology/gene-level isoform annotations do not resolve CLDN18.2 or FGFR2b expression.

Table 3: Tier 1 and Tier 2 candidates. Coarse unordered Tier 1 and Tier 2 nominations from the final frozen ranking, stability outputs, and manual curation. Columns report rank, score, prevalence flag, stability frequency, component values, Wang 2026 class/glyco-outlier context, and the principal caveat. The table is a hypothesis list, not a clinical target list.

rank	gene	tier	primary score	prevalence flag	primary top20 freq	Surf	E	N	R high worse	P	T	wang class	wang glyco outlier rank	principal caveat
1	ITGB4	Tier 1	0.626035	Broad	1.000	0.600000	0.988126	0.995176	0.364564	0.889044	0.789308	Membrane	292	CAVEAT: basal/stratified epithelial (skin keratinocyte/hemidesmosome) normal expression; skin not in organ-risk model so R may under-weight it; preclinical agents only
6	CDH3	Tier 1	0.608164	Broad	1.000	0.600000	0.831911	0.961410	0.312987	0.954444	0.789308	Membrane	NA	established TAA; confirm membranous tumor IHC in pass 2
9	NECTIN2	Tier 1	0.588737	Broad	1.000	0.600000	0.980705	0.988497	0.599629	0.972610	0.789308	Membrane	NA	CAVEAT: purity-adjusted myeloid/DC partial 0.32 p<0.001 + LAMP3+ DC checkpoint literature; closest-to-threshold Tier 1 member; single-cell confirmation needed (user-confirmed Tier 1 2026-05-31)
12	CEACAM5	Tier 1	0.584828	Broad	1.000	0.600000	0.918738	0.936178	0.470501	0.984349	0.702923	ADC/ Antibody	4	apical/normal GI epithelial expression; specificity review
13	JAG1	Tier 1	0.584280	Broad	1.000	0.600000	0.917625	0.813729	0.470501	0.989939	0.936182	Membrane	NA	CAVEAT: dual tumor-epithelial + endothelial/ vascular (angiocrine) expression; most compartment-exposed Tier 1 member; surface-targetable (neutralizing mAbs) but needs single-cell resolution
14	EPCAM	Tier 1	0.580488	Broad	1.000	0.600000	0.964007	0.989239	0.599629	0.840973	0.936182	ADC/ Antibody	22	high normal epithelial expression (preregistered normal-expression penalty)
7	MPZL1	Tier 2	0.605739	Broad	1.000	0.600000	0.974026	0.994434	0.470501	0.964226	0.715131	Membrane	371	needs cell-resolution to confirm tumor-epithelial origin
8	ITGB5	Tier 2	0.595199	Broad	1.000	0.600000	0.959184	0.858627	0.470501	0.947457	0.936182	Membrane	40	high TME + broad expression
10	BST2	Tier 2	0.586798	Broad	1.000	0.800000	0.985158	0.878108	0.364564	0.536333	0.866075	Membrane	NA	membrane localization discordance; hematopoietic risk
11	IFNGR1	Tier 2	0.585684	Broad	1.000	0.600000	0.952876	0.824490	0.470501	0.937954	0.936182	Membrane	141	ubiquitous cytokine receptor, no tumor selectivity
16	ALPG	Tier 2	0.574731	Moderate	0.840	0.600000	0.513544	0.980334	0.085343	0.781163	0.858491	Approved Non Oncology	NA	excellent normal-tissue window but ranking is single-layer dependent
18	DSC2	Tier 2	0.573364	Broad	0.900	0.600000	0.799629	0.803340	0.312987	0.976244	0.789308	Membrane	49	literature reports DOWNregulation in tumor (poor window)
19	TNFRSF11A	Tier 2	0.572773	Broad	0.940	0.600000	0.737291	0.939889	0.290167	0.842929	0.789308	Membrane	NA	weak gastric-specific antibody rationale (denosumab targets ligand)
20	ERBB2	Tier 2	0.568251	Broad	0.340	0.600000	0.968089	0.952134	0.599629	0.901342	0.789308	ADC/ Antibody	232	established HER2 benchmark; NOT forced to Tier 1; robust top50, heterogeneity-consistent
25	CD9	Tier 2	0.560111	Broad	0.020	0.600000	0.991095	0.892022	0.470501	0.923421	0.490751	NA	NA	short extracellular loops limit antibody access
26	LSR	Tier 2	0.559871	Broad	0.040	0.400000	0.989981	0.998516	0.599629	0.954444	0.789308	NA	NA	favorable literature window; tight-junction accessibility caveat
28	CDH17	Tier 2	0.557068	Broad	0.000	0.400000	0.848237	0.871614	0.364564	0.882616	0.936182	Membrane	19	GI-restricted surface-antigen program with clinical-stage CAR-T/ADC context; bulk ranking conservative

Table 3: Tier 1 and Tier 2 candidates. Coarse unordered Tier 1 and Tier 2 nominations from the final frozen ranking, stability outputs, and manual curation. Columns report rank, score, prevalence flag, stability frequency, component values, Wang 2026 class/glyco-outlier context, and the principal caveat. The table is a hypothesis list, not a clinical target list. (continued)

rank	gene	tier	primary score	prevalence flag	primary top20 freq	Surf	E	N	R high worse	P	T	wang class	wang glyco outlier rank	principal caveat
30	TGFBR1	Tier 2	0.553247	Broad	0.000	0.600000	0.860482	0.920594	0.470501	0.951928	0.583426	Experimental	NA	ubiquitous signaling receptor, no tumor selectivity

Table 4: Benchmark-control diagnostics. Positive controls, secondary benchmarks, intracellular negative controls, and TME/off-tumor controls used to assess whether the pipeline behaves plausibly. Control behavior is reported honestly, including the failed aggregate positive-control top50 gate and the cause-level interpretation of misses. Controls were not used to retune weights or rescue individual candidates.

control set	gene	expected	presence status	baseline rank	top20 frequency	top50 frequency	leave one layer rank range	interpretation
positive_control	ERBB2	top_50_in_balanced_or_documented_biological_exception	ranked_in_core_probable_universe	20	0.340000	1.000000	10-46	benchmark_diagnostic_not_weight_tuning
positive_control	CLDN18	top_100_in_balanced_and_isoform_flag_present	ranked_in_core_probable_universe	1474	0.000000	0.000000	886-2154	benchmark_diagnostic_not_weight_tuning
positive_control	FGFR2	top_100_or_subtype_amplification_flag	ranked_in_core_probable_universe	1333	0.000000	0.000000	918-1953	benchmark_diagnostic_not_weight_tuning
positive_control	TACSTD2	ranked_with_normal_risk_flag	ranked_in_core_probable_universe	202	0.000000	0.000000	162-323	benchmark_diagnostic_not_weight_tuning
positive_control	EPCAM	surface_evidence_positive_with_normal_expression_penalty	ranked_in_core_probable_universe	14	1.000000	1.000000	6-38	benchmark_diagnostic_not_weight_tuning
positive_control	CEACAM5	surface_evidence_positive_with_specificity_review	ranked_in_core_probable_universe	12	1.000000	1.000000	8-34	benchmark_diagnostic_not_weight_tuning
positive_control	MET	ranked_or_flagged_for_heterogeneity	ranked_in_core_probable_universe	45	0.000000	0.788000	27-105	benchmark_diagnostic_not_weight_tuning
positive_control	MSLN	surface_evidence_positive_or_documented_coverage_gap	ranked_in_core_probable_universe	158	0.000000	0.000000	48-316	benchmark_diagnostic_not_weight_tuning
secondary_benchmark	CEACAM6	surface_evidence_positive_with_specificity_review	ranked_in_core_probable_universe	57	0.000000	0.244000	29-119	benchmark_diagnostic_not_weight_tuning
secondary_benchmark	ERBB3	ranked_or_documented_if_low_in_STAD	ranked_in_core_probable_universe	101	0.000000	0.000000	19-274	benchmark_diagnostic_not_weight_tuning
secondary_benchmark	GUCY2C	surface_evidence_positive_with_gi_normal_expression_flag	ranked_in_core_probable_universe	345	0.000000	0.000000	132-641	benchmark_diagnostic_not_weight_tuning
secondary_benchmark	SLC44A4	topology_and_ecd_review_required	ranked_in_core_probable_universe	427	0.000000	0.000000	241-953	benchmark_diagnostic_not_weight_tuning
secondary_benchmark	NECTIN4	ranked_or_documented_if_expression_low_or_risk_high	ranked_in_core_probable_universe	389	0.000000	0.000000	202-709	benchmark_diagnostic_not_weight_tuning
negative_control	ACTB	not_top_100_and_not_surfaceome_core	not_in_core_probable_universe	—	—	—	—	benchmark_diagnostic_not_weight_tuning
negative_control	GAPDH	not_top_100_and_not_surfaceome_core	not_in_core_probable_universe	—	—	—	—	benchmark_diagnostic_not_weight_tuning
negative_control	H3C1	not_top_100_and_not_surfaceome_core	not_in_core_probable_universe	—	—	—	—	benchmark_diagnostic_not_weight_tuning
negative_control	TOMM20	not_top_100_and_not_surfaceome_core	not_in_core_probable_universe	—	—	—	—	benchmark_diagnostic_not_weight_tuning
negative_control	CALR	not_tier_1_without_context_specific_surface_evidence	not_in_core_probable_universe	—	—	—	—	benchmark_diagnostic_not_weight_tuning
negative_control	ALB	not_top_100_and_not_cell_surface_target	not_in_core_probable_universe	—	—	—	—	benchmark_diagnostic_not_weight_tuning
tme_or_off_tumor_penalty_control	PTPRC	surface_evidence_allowed_but_not_tier_1_for_epithelial_tumor_targeting	ranked_in_core_probable_universe	636	0.000000	0.000000	290-1230	benchmark_diagnostic_not_weight_tuning

Table 4: Benchmark-control diagnostics. Positive controls, secondary benchmarks, intracellular negative controls, and TME/off-tumor controls used to assess whether the pipeline behaves plausibly. Control behavior is reported honestly, including the failed aggregate positive-control top50 gate and the cause-level interpretation of misses. Controls were not used to retune weights or rescue individual candidates. (continued)

control set	**gene**	**expected**	**presence status**	**baseline rank**	**top20 frequency**	**top50 frequency**	**leave one layer rank range**	**interpretation**
tme_or_off_tumor_penalty_control	PECAM1	surface_evidence_allowed_but_not_tier_1_for_epithelial_tumor_targeting	ranked_in_core_probable_universe	15	1.000000	1.000000	15-39	benchmark_diagnostic_not_weight_tuning